

View results

Respondent

63

Anonymous

205:11

Time to complete

A. General information about the respondent

1. * Country

Argentina



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Under Secretariat of Science and Technology

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://www.argentina.gob.ar/ciencia>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

Contributes to the design and implementation of the S&T policy and the dissemination of science and technology.

7. * Full name of officially designated contact person(s) who completed this survey

María Guillermina D'Onofrio

8. * Position of officially designated contact person(s) who completed this survey

National Director of Information and Studies

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Paula Nahirñak

11. Brief description of the consultation process established for the preparation of the report

Information was collected from official web sites and enquiries made to relevant entities and organizations, focused on the institutions belonging to the National Science, Technology and Innovation System (SNCTI) as described by Law 25.467. In addition, a review of programs and funding of the Under Secretariat of Science and Technology (SSCYT), the National Agency for the Promotion of Research, Technological Development and Innovation (Agencia I+D+i) and the National Scientific and Technical Research Council (CONICET) was conducted.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

National Scientific and Technical Research Council (CONICET)

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<https://www.conicet.gov.ar/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

☒ Public

☐ Private

☐ Civil Society

☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN
SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒ Yes

☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The former Ministry of Science, Technology and Innovation (MINCYT) (at present Secretariat of Innovation, Science and Technology), the National Scientific and Technical Research Council (CONICET) and the Ministry of Foreign Affairs shared the news of the adoption in their webpages. The Recommendation was also posted at the website of several National Universities, including the University of Buenos Aires and the National University of Cordoba, the two largest universities by student enrollment in Argentina. Please see links in Annex.

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Spanish

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.



Yes



No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Please find below some activities conducted during the last years: The former Ministry of Science, Technology and Innovation (at present Secretariat of Innovation, Science and Technology) is a central entity for raising awareness about Open Science. Among its actions in the last few years we can mention: Series of Workshops and Conversations on Open and Citizen Science, held between 2018 and 2019, which included various activities such as: Workshop on Citizen Science (2018), Workshop on Open Research Data (2018), Workshop on Open Hardware (2018), Workshop: Collaborative Citizen Science Clinic (2019), and Evaluation from Open Science Perspectives: Proposal of Indicator Frameworks (2019). Presentation of the Diagnosis and Lineaments for an Open Science Policy in Argentina (2022). Organización del 1st National Meeting on Citizen Science (2023). The former Ministry of Education (at present Secretariat of Education) through the National Inter-University Board (CIN) and national universities are also promoters of open science. Their actions include the following: International Conference of Public Universities (2022) (Open science and citizen participation in knowledge and the scientific agenda was a pillar). Argentine Meeting on Open and Citizen Science (launched in 2017 as a workshop): Workshop on Open and Citizen Science (2017), 2nd Meeting on Open and Citizen Science (2018), and 3rd Argentine Meeting on Open and Citizen Science (2021). International Conference on Research Data (2024) hosted by the Research Data Alliance at the National University of Rosario. International Conference on Research Data (2024) by the Research Data Alliance at the National University Rosario. Lastly, of note, a number of ministries, science and technology institutions and universities actively participate in international proposals such as Citizen Science Month and Open Access Week. Please, see more activities related to Open Science in point 4.1.

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

The goals and principles of Law 25.467 of Science, Technology and Innovation, approved in 2001, match a number of goals and principles of the UNESCO Recommendation. Among its goals, the Law includes the social utilization of knowledge; the dissemination, transfer, coordination, and spreading of such knowledge; contributing to social well-being; ensuring equal opportunities; promoting international cooperation actions. And among its principles: respect for the dignity of individuals; non-discrimination based on physical condition, health, history, and genetic data; protection of vulnerable groups; care and protection of the environment and biodiversity of all species. Even if some calls for public funding may not explicitly list these principles, Law 25.467 is a framework that governs other laws and programs in the public sector: Law 26.899 of Open Access Institutional Digital Repositories, National Science, Technology and Innovation Plan 2030, National Citizen Science Program, National Systems, Federal Program "Equipping Science", etc. At present, CONICET is preparing a call for research projects, and they are considering how best to include a number of values and principles of open science both in the terms of reference and in the evaluation process, in addition to the open science values and principles already being applied.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

25. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

The National STI Plan 2030 that was passed into law in October 2023 (Law 27.738). The former Minister of Science, Technology and Innovation was responsible for developing the plan, and the provinces were involved in its preparation. The Plan seeks to address 10 national challenges (aligned with the UN SDG) with 4 different types of agendas (strategic, territorial, crosscutting and institutional change agendas). Elements of Open Science appear in these agendas, mainly in the section "Citizenship and Scientific Culture" where emphasis is placed on Open Access and Citizen Science. Link: National Science, Technology and Innovation Plan 2030. A Guideline for implementing the Plan is under development.

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☒ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

☒ Yes

☐ Partly

☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this

link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

In 2013 Argentina passed the Law 26.899 of Open Access Institutional Digital Repositories. This Law was implemented in 2016 by Resolution 753 - E/2016 of the Ministry of Science, Technology and Innovation. This Law establishes a regulatory framework for the Open Access policy at the national level. It mandates that public bodies and institutions within the National Science, Technology and Innovation System (SNTI), receiving funding from the National Government, develop digital institutional repositories in open access. These repositories, whether individually owned or shared, must comply with international interoperability standards. They are intended to collect scientific and technological results from work, training, and/or projects wholly or partially funded by public funds. All such results must be deposited in these repositories. Furthermore, the Law requires the establishment of policies to ensure public access to primary research data. This will be achieved through open access digital institutional repositories or national portals for large instruments and database systems. It also mandates institutional policies for the long-term management and preservation of these resources.

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).



Yes



Under development



No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Several programs of the Under Secretariat of Science and Technology of the Secretariat of Innovation, Science and Technology promote open science or a number of elements of open science: The National System of Digital Repositories (SNRD), created in 2011, is an initiative that aims at developing an interoperable network of S&T digital repositories by establishing common policies, standards and protocols for all members of the National S&T System. The initiative partially funds projects for creating or strengthening digital repositories; the production of these repositories is harvested by the National Digital Repositories System portal. This instrument promotes open access. Connected with this policy, the Open Access Primary Data Portal for Science and Technology in Argentina (DACyTAr) was created in 2020. Allows users to search and access primary research data in open access through digital institutional repositories included in the SNRD in a centralized manner. The National Citizen Science Program was created in 2022 by Ministerial Resolution 620/2022 and aims to promote Citizen Science all over Argentina by contributing to the development of CC initiatives, raising awareness, promoting joint construction of knowledge and active participation of societal actors in the various stages of CC initiatives. This instrument promotes citizen science as an element of open science. National Systems promote an efficient use of large equipment and facilities. The National Science, Technology and Innovation System (SNCTI) institutions that enrol their equipment and instruments in the Program are required to make them available to third-party users for 20% of the available time. This initiative, created in 2008, promotes a shared use of scientific infrastructure, which is a step in the right direction to achieve open infrastructures in line with open science.

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☒ Yes
- ☐ Under development
- ☐ No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

An Advisory Open and Citizen Science Committee was set up in 2021 by Ministerial Resolution 132/2021, made up of members from various disciplines and institutions (MINCYT, CONICET, Universities and other S&T organizations). In 2022 the Committee issued a Diagnosis and Lineaments for an Open Science Policy in Argentina, including a number of recommendations. This document provides elements to consider as a baseline for a National Open Science Strategy, the roadmap of which is planned to be developed shortly.

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

Law 26.899 promotes the implementation of open access policy by the National Science, Technology and Innovation System (SNCTI) institutions. Among the institutions, public and private, under the scope of the Law, 18 have already implemented such policies and 13 have draft versions (in addition, we registered 18 institutions that have some form of Open Access mandate). Likewise, 81 institutions have at least one digital repository in operation. Considering only public universities and the main national science and technology organizations, our survey shows that out of 91 institutions, 51.6% have at least one active digital repository, while 29.7% do not have a repository but offer journals, publications, and/or open data on their websites. From these results, we infer that there is still work to be done in the field of Open Access. Likewise, a number of universities have included elements of open science, mainly in connection to open access, to their strategic plans, for example Strategic Plan 2022-2026 of Universidad Nacional de La Plata, Participatory Strategic Plan 2030 of Universidad Nacional de Mar del Plata, Institutional Project 2016-2022 of Universidad Nacional de San Antonio de Areco and Strategic Plan and Institutional Development Plan 2016-2020 of Universidad Nacional de La Pampa.

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.



Yes



Under development



No

36. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

Specific funding mechanisms include: The National Citizen Science Program aims at funding Argentine Citizen Science programs and projects, from the ideation stage to improvement and scaling up (among other objectives, such as mapping and monitoring CS programs and projects, encouraging the participation of civil society and promoting training, research and diagnoses related to CS). A call was launched in 2023. The National System of Digital Repositories aims to provide financial support for creating and/or strengthening digital repositories and for the training of human resources. National Systems aims to support institutions for upgrading equipment, purchasing supplementary equipment and Human Resources training, under the vision of shared use of instruments. Federal Program "Equipping Science" awarded funds to beneficiaries to purchase medium size and large equipment. Beneficiary institutions undertook to enrol their instruments in the National Systems program and comply with the Open Access Law. The most recent evaluation of applications took place in 2023.

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.



Yes



No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

Some funding awarded by the Under Secretariat of Science and Technology and the Agencia I+D+i usually requires beneficiaries to comply with the Open Access and Digital Repositories Law, as can be seen in the 2022 and 2023 Calls for applications for Science and Technology research projects, the National Systems Program and the Federal Program "Equipping Science". Particularly for funds awarded by the National Digital Repositories System, the beneficiary institution must have in place an Institutional Open Access Policy in line with the provisions of Law 26.899 and its implementation regulation. A number of national programs require collaboration between researchers and institutions and between them and local societal actors. These include Impacting Program for R&D projects that provide a solutions to public interest problems; Interinstitutional Centers in Strategic Subjects and Scientific and Technological Facilities Program that funds the creation of centers on strategic issues and large facilities, equipment and infrastructure projects (in both programs, at least 3 S&T institutions are involved and local and provincial entities may also be part of the project). At institutional level, several examples include Research and Development Projects 2025 – Universidad Nacional de La Plata that requires beneficiaries to comply with Law 26.899 and university open access institutional policies; the rules for the UBACYT grants of Universidad de Buenos Aires that mandates compliance with institutional repository requirements; the terms of the Call for Technological and Social Linkage Projects 2024 of the Universidad Nacional de los Comechingones that require projects to include specific territorial stakeholders for eligibility thus promoting collaboration with local societal actors and local communities.

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

40. 2.8 cont. **If yes**, please provide details and links as relevant.

CONICET is currently developing efforts to foster equitable public-private partnerships on open science; however no document has yet been published online.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

☐ Yes

☒ Under development

☐ No

42. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

Specific indicators on open science are under development. Argentina has an Integrated S&T Indicators System under the scope of the National Directorate of Information and Studies. Surveys about R&D and innovation are regularly conducted at academic, governmental and company level. The National Directorate of Information and Studies is at present developing an Open Science Monitor; as a first step towards that project, a Monitor for Law 26.899 is currently underway. Of note, in 2023 the first National Mapping of Citizen Science Initiatives was conducted. This provided information on the number of projects, actors involved, geographical distribution, etc. of Citizen Science initiatives.

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

43. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

Latest official data revealed that 0.60 % of GDP was dedicated to research and development expenditure in 2023. See at <https://app.powerbi.com/view?r=eyJrJoiOTdhZjBNDUtyYmY0YS00MjFmLWFiYTgtNDNINjg1MmM3OWUwliwidCI6ImNmNTFjNGM0LTkOTktNGYzYi1iZmE3LTQ3ZGEyOTk2YjIzZCIsImMiOiR9>.

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

Latest official data corresponds to the Report on Access and Use of Information and Communication Technologies from the Permanent Household Survey, fourth quarter 2023. The Report informed that 93.4% of households of 31 urban agglomerations surveyed have Internet access.

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.



Yes



No

46. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

InnovaRed, a project of InnovaT Foundation, a technological linkage unit of CONICET. This NREN connects various organizations within the scientific, technological and educational system, including universities, science and technology entities, ministries, observatories, foundations, etc. InnovaRed is also part of RedCLARA, Advanced Internet Network of Latin America. The University Interconnection Networks Association should also be mentioned. With more than 59 universities enrolled, this Association provides connectivity to its members.

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☐ Yes
- ☒ Under development
- ☐ No

48. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☒ community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- ☒ platforms for exchanges and co-creation of knowledge between scientists and society
- ☒ community-based monitoring and information systems to complement national, regional and global data and information systems

49. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

There is a number of open science infrastructures, including the following: High performance computing: Clementina XXI: Supercomputer installed at the National Meteorological Service (SMN), which is available for use to the entire science and technology community in the country. At present, the Accelerated Computing Projects Initiative (IPAC) is currently open. This Initiative provides the entire National Science, Technology and Innovation System (SNCTI) with competitive open access to the large scale computing resources of Clementina. The National System of High Performance Computing, under the Under Secretariat's jurisdiction, provides a framework for shared equipment use. Its 28 affiliated centers commit to making 20% of their computing time available for shared access. Large size equipment: Over 600 large size equipment are enrolled in the National Systems program, thus being available for use by the scientific community and other sectors that require them. Repositories for publications: The National Digital Repositories System (SNRD) is an interoperable network of digital repositories for science and technology. Currently 79 repositories are enrolled in the System, with 52 being harvested in the portal. The System uses OAI-PMH (Open Archives Initiative Protocol for Metadata Harvesting) and its standards are based on international standards. Open Data: Besides the above mentioned DACyTar portal, the following National Science, Technology and Innovation System (SNCTI) institutions open their data and provide services in their relevant fields: Open Data of the National Meteorological Service; Mining Environmental Geological Information System (SIGAM) of the Argentine Geological Mining System; Hydrographic Services of the Naval Hydrographic Service; Geographic Resources and Database of the National Water Information System (SNIH) of the National Water Institute; INTAGeo and Agroclimatic Information of the National Institute of Agricultural Technology; Geographic Services and Data of the National Geographic Institute; Geospatial Information Portal of the National Commission on Space Activities; Biodiversity Information System of the National Parks Administration; Research Data Repository of CONICET; Spatial Data Infrastructure of Universidad Autónoma de Entre Ríos. Museums of Science and Scientific Parks: Several institutions within the National Science, Technology and Innovation System (SNCTI) in Argentina feature thematic museums that engage the public with science and technology; notably, the Cultural Science Center (C3) under the jurisdiction of the Under Secretariat of Science and Technology, that facilitates access to scientific culture for society at large; and TECNOPOLIS, a science, technology, industry and art exhibition, the largest in Latin America. Both are open to the general public free of charge. Archives: Argentina boasts several archives, documentation centers and other organizations with documentary collections, including the National Historic Documentation System, that shares with the community information about organizations with such collections (see the entity search engine). Platforms for co-creating knowledge between scientists and society, community-based information and monitoring systems: Several Citizen Science projects were supported by computer platforms (see https://www.argentina.gob.ar/sites/default/files/ciencia_ciudadana-es-3ra-edicion.pdf). These projects exemplify the various ways in which scientists and society collaborate to generate knowledge.

50. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☒ Yes

☐ No

51. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Argentina is a member of The Latin American Network for Open Science (LA Referencia). LA Referencia is the result of a cooperation agreement signed in 2012. It supports national Open Access strategies in Latin America and Spain. Through its platform it shares scientific production, in line with interoperability standards. The National System of Digital Repositories is one of the 12 nodes that make it up.

52. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒ Yes

☐ No

53. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Argentina is a member of The Latin American Network for Open Science (LA Referencia). LA Referencia is the result of a cooperation agreement signed in 2012. It supports national Open Access strategies in Latin America and Spain. Through its platform it shares scientific production, in line with interoperability standards. The National System of Digital Repositories is one of the 12 nodes that make it up.

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

54. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

55. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Between 2021 and 2024 around 30 activities related to elements of Open Science (Open Access, Citizen Science, Open Educational Resources, among others) were organized by over 20 institutions, mainly universities (please see details in the Annex).

56. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

57. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

It is expected that officials and policy makers will be trained once a National Open Science Policy has been outlined in Argentina.

58. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☒ Yes

☐ No

59. 4.3 cont. **If yes**, please provide more information and links as relevant.

Aspects related to open science have been incorporated in curriculums, though this is not the result of any central planning but the individual initiative of institutions. This includes the library science and related studies at national universities: Licenciatura in Library Science and Information Science, University of Buenos Aires; Licenciatura in Library Science and Information Science, National University of La Plata; Licenciatura in Library Science, National University of Litoral; Licenciatura in Library Science and Documentation, National University of Mar del Plata; Licenciatura in Library Science and Documentation, National University of Córdoba.

60. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☒ Yes
- ☐ Partly
- ☐ No

61. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

62. 4.5 cont. **If yes**, please provide more information and links as relevant.

At national level: Educ.ar: educational portal of the Secretariat of Education under the Ministry of Human Capital; and MetaDocencia: an initiative supported by NASA. Among the initiatives at National Science, Technology and Innovation System (SNCTI) organizations: Toolkit, National University of General San Martín. Collaborative Resources Repository, National University of Luján. Open Educational Resource Repository, National University of Patagonia Austral. Ansenzuza, National University of Córdoba. Software 2Mp, National Commission on Space Activities (CONAE). Open Educational Resources UNLP, National University of La Plata. At national level: Educ.ar: educational portal of the Secretariat of Education under the Ministry of Human Capital; and MetaDocencia: an initiative supported by NASA. Among the initiatives at National Science, Technology and Innovation System (SNCTI) organizations: Toolkit, National University of General San Martín. Collaborative Resources Repository, National University of Luján. Open Educational Resource Repository, National University of Patagonia Austral. Ansenzuza, National University of Córdoba. Software 2Mp, National Commission on Space Activities (CONAE). Open Educational Resources UNLP, National University of La Plata. At national level: Science Culture Projects, funded under the National Science Popularization Program. These projects aim to demonstrate the social value of science and technology through associative and interinstitutional initiatives, with a federal perspective. The results of the most recent edition (2023) are available here. Scientists go to School: this initiative promotes collaborative work between teachers and the scientific community to formulate empirical projects aimed at enriching the lessons of science and technology-related subjects. Argentine Science and Technology Week: annual meeting between the scientific community and the public for the popularization of science, technology, and innovation. During one week, simultaneous public communication, outreach, and dissemination activities take place across the country. Most recent edition: 2023. As mentioned in section 3.4, the country has museums and parks that bring scientific knowledge closer to the public, such as TECNOPOLIS and the Cultural Center of Science (C3). The initiative At Open Science (at the C3) engages different audiences in scientific projects. The National Science, Technology and Innovation System (SNCTI) organizations also have a number of initiatives to support science communication in line with the value and principles of open science, such as: ConCIENCIAR, National University of Villa María. UNNE Science goes to your school, National University of Nordeste (UNNE). San Juan Science and Technology Festival "Create with Science", National University of San Juan and CONICET. Science Fair, National University of Salta - CONICET. Microscience, National Fisheries Research and Development Institute. Open Constituyentes Atomic Center (CAC) - National Atomic Energy Commission 9th International Meeting of Public Science Communication, National University of San Luis. Muestra Anual de Arte y Ciencia, National University of Comechingones. Open Science in Pandemic Times podcast y Citizen Science Dissemination podcast, National University of Córdoba. Open Access in movement Blog, National University of La Plata. Interviews on Open Science, Scientific Research Commission of the Province of Buenos Aires (CIC).

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

63. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒ Yes

☐ No

64. 5.1 cont. **If yes**, please provide more information and links as relevant.

The National Science Technology and Innovation Plan includes an Agenda of Institutional Change, an item of which is to design an evaluation model with indicators that consider open access relevant science and technology production. CONICET has strengthened its evaluation criteria by incorporating elements of open science that are transversal to all areas of knowledge in the different stages of evaluation. The Institution formally adhered to the Declaration on Research Assessment (D.O.R.A.) in 2022. In addition, it is working to incorporate these principles in various assessment stages (entry, continuity, promotion, funding and the grant program). Working documents are also prepared for the advisory bodies that further develop these concepts. See: Assessment Guidelines for Management of Researchers' Careers and Assessment Guidelines for Technological Products and Activities.

65. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☒ Yes

☐ No

66. 5.2 cont. **If yes**, please provide details and links as relevant.

The implementation of Plan 2030 is underway and we expect to have a National Open Science Policy that will thoroughly consider the incorporation of open science practices in career evaluation and progression systems. CONICET is also working along these lines as mentioned above in section 5.1.

67. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025?
(ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

Yes



No

68. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

National Data Visualization Competition "Contar con Datos": Hosted by the SSCYT and the Data Science Lab of the University de San Andrés, it aims to promote the use of scientific data visualization, encourage engagement with public communication of science, inspire interest in data journalism, and raise visibility for researchers working in the field of Data Science across the country. RIA Awards - National Technological University: This initiative recognizes the most visited works, the most searched authors and the authors with the largest number of digital objects deposited at its institutional repository, and the regional schools with the highest percentage of uploads.

69. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

70. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

☐ YES, at the national level

☐ YES, at the institutional level

☒ YES, at both the national and institutional levels

☐ No

71. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☒ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☒ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☒ promoting open innovation practices
- ☐ Other

72. 6.1. cont. **If yes**, please provide more information and links as relevant

promotion of pre-prints: Yes, at the national level: Article 5 of Law 26.899 sets out that: "researchers, technologists, professors, postdoctoral fellows, and master's and doctoral students whose research is funded with public money must deposit, or explicitly authorize the deposit of, a copy of the final version of their scientific and technological work—whether it has been published, accepted for publication, or undergone an approval process by a relevant authority—into their institution's open-access digital repositories. This must be done within six (6) months of its official publication or approval." participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas. Yes, at the national level: As part of the National Climate Change Adaptation and Mitigation Plan, the former Ministry of Environment and Sustainable Development set out to initiate a participatory process that recognize the Indigenous Peoples' inherent right to be consulted and award their Free, Prior and Informed Consent (FPIC) regarding any measure or project that affects them. Thus, four regional workshops were held with Indigenous community leaders, called "Intercultural Dialogues," aimed at identifying the main risks and impacts of climate change, taking into account their worldview, and thus facilitating coordination with national ministries. The report can be accessed here. Along the same lines, the former Ministry of Environment, also consulted with indigenous peoples for the design of the National Action Plan for Forests and Climate Change (PANByCC), thus respecting the knowledge and rights of local communities. promoting open innovation practices. Two initiatives led by national organizations are highlighted: Ibero-American Diploma in Open Innovation, Secretariat of Knowledge Economy under the Ministry of Economy, and the National Technological University: Aimed at professionals, this program aims to strengthen the concept of open innovation in the context of digital and green transformation. Virtual Course on Open Innovation, Ministry of Productive Development and others: A training program designed to raise awareness and promote the benefits of open innovation in relation to production, industry, and organizations.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

73. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Argentina is a member of LA Referencia and, through this organisation, is also a member of the Confederation of Open Access Repositories (COAR). In 2023 LA Referencia signed a cooperation agreement with RedClara and LATindex for the purpose of consolidating a Regional Open Science Ecosystem. More information here. In 2021 LA Referencia also signed a MoU with ASREN, WACREN and UbuntuNet Alliance, African regional research and education networks, to promote open science activities, sharing information on an ongoing basis and the potential adoption of the LA Referencia open source discovery software in Africa (see here). The collaboration is fostered by COAR and with support from the pan-European GÉANT network, OpenAIRE and the EU co-funded AfricaConnect3 project. Under Pampa Azul (an inter-ministerial initiative) Argentina participates in CoastCarb: Coastal ecosystem carbon balance in times of rapid glacier melt. This project brings together over 100 experts from 10 countries to create a knowledge information system with open access data portal and produce dynamic ecosystem models for fjordic and estuarine environments. Argentina cooperates with the EU and other countries in the initiative 'Open Science: Innovations, Challenges and Projections for the Future based on an Inter-regional Dialogue'. CONICET and Universidad de Antioquía of Colombia signed a letter of undertaking in 2019 to conduct a study on the status of open access and APCs in Argentina. The report was released in 2022. Mercosur - Special Meeting on Science and Technology (RECyT). An Open Science Group was set up by the end of 2022, and met virtually during 2023/24. The Group is working on a Recommendation on Open Science.

74. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

75. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

76. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☒ Yes

☐ No

77. 7.3 cont. **If yes**, please provide links as relevant and more information.

In the context of Mercosur-RECyT the Open Science Group has been discussing the creation of funding mechanisms for regional multi-center projects that seek to reproduce key findings as well as the establishment of regional instruments to fund open science initiatives, with focus on the collaboration between member countries.

Part 8. GENERAL CONSIDERATIONS

78. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

79. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org

No.